

MoRE: Mixture-of-Experts for Multi-Physics Robotic World Models

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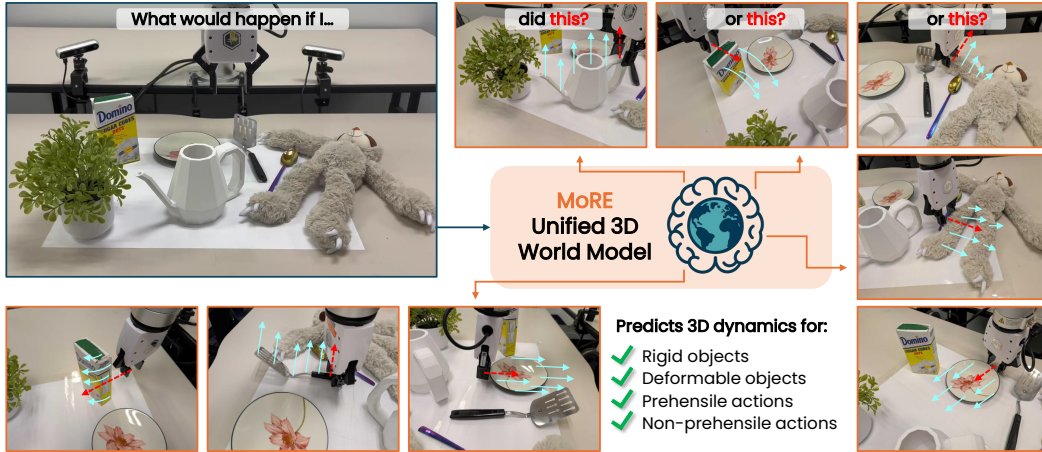


Figure 1: **MoRE** predicts action-conditioned 3D dynamics by routing tool-object interactions to specialized motion experts. This unified MoE model handles both rigid and deformable objects, as well as both prehensile and non-prehensile contact. Trained on mixed simulation and real data, MoRE supports real-time prediction at 20 FPS and generalizes across tasks, materials, and objects.

Abstract: A robotic world model should predict how objects evolve under robot actions and physical dynamics. While recent 3D world models have shown strong predictive capabilities, they still struggle with the heterogeneous dynamics arising from diverse object interactions. To address this in a unified framework, we decompose robotic interactions into distinct physical regimes and assign each to a specialized motion expert. We introduce MoRE, a 3D world model with a mixture of motion experts organized around rigid, deformable, and prehensile dynamics. This design enables a single model to capture diverse physical behaviors while preserving regime-specific priors. Trained on mixed simulation and real-world data, MoRE integrates with Model Predictive Control (MPC) to support diverse manipulation tasks. Experiments show that MoRE improves prediction accuracy across physics regimes, leading to stronger real-world MPC performance.

1 Introduction

Robots that operate intelligently in the physical world should be able to predict how the environment evolves under their actions. Such an ability is fundamental for planning, control, and long-horizon decision making [1, 2, 3, 4, 5, 6, 7], and is especially crucial in contact-rich real-world manipulation [8, 9, 10, 11, 12]. For instance, a house-cleaning robot must anticipate how cloth moves when wiping a surface, while a cooking robot must predict how ingredients deform under tool interactions.

Building such a dynamics model is challenging because real-world manipulation involves heterogeneous physical behaviors, complex contacts, and diverse interaction mechanisms. To capture these effects, a robotic world model must be accurate, physically plausible, 3D-aware, and grounded in robot-object interactions. It should generalize across physical regimes, from rigid objects to deformable materials, and across manipulation modes, including prehensile grasping and non-prehensile interactions with rigid or soft tools. In short, we need models that are geometry-grounded, multi-physics, multi-tool, and contact-rich.

Unfortunately, existing approaches address only parts of this problem. Physics-based methods [13, 14, 17] leverage differentiable simulators and can support long-horizon roll-outs, but they often require heavy system identification and expensive optimization. In

contrast, learning-based dynamics models [18, 19, 20, 21, 22, 23, 24, 25, 26, 11] avoid explicit system identification by learning dynamics directly from data, yet most are designed for a specific material type, tool, or task, limiting their ability to generalize across heterogeneous manipulation regimes. More recent generative world models [6, 27, 2, 28] offer greater scalability and broader scene coverage, but they are primarily optimized for visual prediction and often lack the accurate 3D structure and action grounding needed for real robotic execution. This leaves a gap between scalable prediction models and the physically grounded, controllable dynamics required for robot manipulation.

We aim to close this gap with MoRE, a scalable robotic world model that predicts future 3D point trajectories from a short history of object-tool interaction and the current robot action. Our key insight is that scaling alone is insufficient: to generalize across diverse robotic dynamics, scaling must be paired with a physics-informed mixture of motion experts. On the data side, we scale across simulation and real-world settings, covering diverse object geometries, materials, contact patterns, tasks, and tools. On the model side, we avoid forcing all dynamics into a single monolithic predictor. Instead, inspired by interaction-centric dynamics modeling [24, 20, 12, 11, 29] and Mixture-of-Experts (MoE) architectures [30, 31], MoRE decomposes robotic dynamics into specialized motion experts aligned with physical regimes and interaction types, such as rigid/non-rigid and prehensile/non-prehensile. These experts share a common encoder for scalable representation learning, while each expert incorporates regime-specific inductive biases and constraints, such as rigidity preservation. A learned gating module routes each interaction to the appropriate expert based on short motion history, allowing the model to infer material properties and interaction modes directly from observation.

We evaluate MoRE in both simulation and real-world manipulation, covering pushing and grasping with rigid and deformable objects. Trained on a large-scale interaction dataset, our model runs at 20 FPS for model inference and integrates naturally with tuning-free MPC for closed-loop control. Across dynamics benchmarks and real robot tasks, MoRE consistently outperforms prior methods.

In summary, we make three contributions. First, we introduce MoRE, a 3D robotic world model for contact-rich dynamics prediction. Second, we design MoRE around a Mixture-of-Motion Experts architecture that explicitly captures heterogeneous physical regimes and interaction types. Third, we demonstrate that MoRE enables accurate real-time prediction and effective closed-loop control across both simulated and real-world robotic manipulation tasks.

Table 1: **Comparison of robotic manipulation methods** across key capabilities: feed-forward inference, multi-physics dynamics (different material properties of manipulated objects, e.g., rigid vs. deformable), and multi-tool interaction (different end-effector geometries and contact interfaces), as well as 3D-awareness. ParticleFormer shows partial multi-tool support but is trained per scene. PointWorld relies on implicit dynamics while we explicitly route interactions to physics-aligned experts.

Method	Feed-forward	Multi-phys.	Multi-tool	3D
PhysTwin [13]	✗	✗	✗	✓
PhysGaussian [14]	✗	✗	✗	✓
AdaptiGraph [12]	✗	✗	✗	✓
ParticleFormer [15]	✓	✓	✓ [†]	✓
Points2Plans [16]	✓	✗	✗	✓
RoboDreamer [2]	✓	✗	✗	✗
DINO-WM [1]	✓	✓	✓	✗
PointWorld [3]	✓	✓	✓	✓
Ours	✓	✓	✓	✓

2 Related Work

Existing robotic dynamics models can be broadly divided into three categories: physics-based simulation methods, learning-based dynamics models, and foundation robotic world models. We compare our approach with representative prior work across these categories in Table 1.

Physics-Based Dynamics Models Physics-based dynamics models [32, 33, 17] predict manipulation outcomes by explicitly simulating physical interactions. For rigid bodies [34, 35, 36], common formulations use state-based dynamics on the SE(3) manifold with contact constraints, implemented in rigid-body simulators [37, 38, 32, 39, 40]. For deformable and soft objects, simulators typically rely on continuum or particle-based representations, such as MPM [14, 41, 42, 43, 44, 45], PBD [46, 47, 48], or spring-mass systems [49, 13], to capture material-dependent deformation. Recent hybrid pipelines [14, 50, 51, 52] combine reconstruction with physics-informed simulation to build digital twins for future prediction. These approaches provide strong physical priors and support long-horizon rollouts when parameters are accurate, but often depend on system identification and expensive test-time optimization to match observations [13, 14]. Moreover, they typically require complete state and detailed geometry as input, making them suffer under partial observations, sensor noise, or unknown materials. In contrast, our method performs implicit system identification from interaction history, avoiding test-time optimization and explicit parameter identification.

Learning-Based Dynamics Models Learning-based dynamics models [18, 19, 20, 21, 22, 23] bypass explicit simulation by learning action-conditioned state transitions from data. Early work represents objects and interactions using graphs [20, 53], particles [11, 54, 20], or GNNs [24, 25, 26], while more recent approaches adopt transformer-style architectures to improve capacity and scalability for contact-rich interactions. These models enable efficient inference and strong performance in specific regimes, operating directly from observations. However, those models remain *narrow*: they are trained on category or task-specific datasets and implicitly assume fixed materials, tools, or interaction modes. This often leads to overfitting and limited generalization ability. Our model is also learning-based, but is explicitly designed to support diverse materials, physical properties, object geometries, and action-conditioned interactions within a unified framework.

Foundation Robotic World Models Motivated by the success of large visual foundation models [5, 55, 56], recent robotic world models can be broadly grouped into 2D and 3D approaches. 2D methods [6, 29, 57, 58, 27] leverage pretrained visual features or video generative models for prediction and planning; for example, DINO-WM [1] plans in a DINOv2 [56] feature space, while RoboDreamer [2] synthesizes future videos with diffusion models [59]. However, these methods are not explicitly grounded in 3D geometry or contact interactions. 3D methods instead predict scene evolution in geometric space. PointWorld [3], for example, predicts action-conditioned 3D point flow from scene point clouds and robot motion. While promising, existing 3D approaches do not explicitly model the heterogeneous dynamics of robotic interactions. In contrast, our method uses learned gating to route interactions to regime-specific motion experts, enabling more physically faithful prediction across diverse manipulation scenarios.

3 Method

3.1 Problem Formulation

We model a robotic *3D world model* as a dynamical system that predicts future scene evolution under robot actions and physical interactions. Let $\mathbf{s}_t \in \mathcal{S}$ denote the *current* scene state and $\mathbf{a}_t \in \mathcal{A}$ the *current* robot action, where $\mathcal{S}$ and $\mathcal{A}$ are geometry-based point-cloud spaces defined below. Given the current state-action pair $(\mathbf{s}_t, \mathbf{a}_t)$ together with a finite history of past states and actions $\{\mathbf{s}_{t-H':t-1}, \mathbf{a}_{t-H':t-1}\}$, the goal is to predict future scene states over a horizon of H steps.

We parameterize the world model as

$$\mathbf{s}_{t+1:t+H} = f_{\theta}(\mathbf{s}_t, \mathbf{a}_t, \mathbf{s}_{t-H':t-1}, \mathbf{a}_{t-H':t-1}). \quad (1)$$

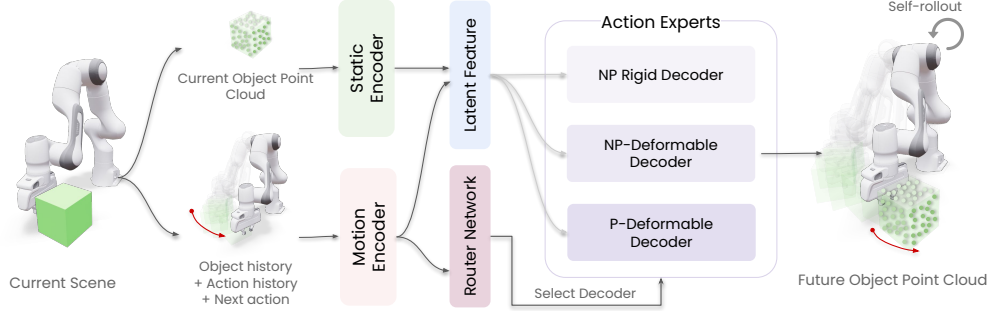


Figure 2: **Overview of MoRE.** Given the current and historical object states and actions, represented as point clouds, we first encode the scene and actor inputs into point tokens using two encoders. An explicit gating network then selects among specialized motion experts, each tailored to a distinct physical regime, to predict the next object state in 3D. P = prehensile, NP = non-prehensile.

We represent each scene state as a 3D point cloud, $\mathbf{s}_t \in \mathcal{S} := \mathbb{R}^{N_S \times 3}$ with N_S being the number of scene points. Crucially, we design the action space $\mathcal{A}$ in a geometry-centric manner: unlike most prior work that conditions dynamics on joint commands, end-effector poses, or sparse keypoints [60, 61, 62], we represent each action by the full 3D state of the *tool* as a point cloud, $\mathbf{a}_t \in \mathcal{A} := \mathbb{R}^{N_A \times 3}$. Specifically, low-level robot controls are mapped via forward kinematics to $\mathbf{a}_t$, where the tool is any directly actuated geometry by the embodiment (e.g., grippers, hands, cookware, or daily tools), as long as its state can be computed via forward kinematics. This representation decouples the world model from embodiment-specific kinematics, enabling cross-embodiment transfer and multi-tool generalization across diverse manipulation tasks. Conditioned on the current and historical scene-action point clouds, the model predicts future scene point clouds $\mathbf{s}_{t+1:t+H}$, capturing rigid-body motion, deformation, and contact-induced dynamics.

3.2 Mixture-of-Experts Dynamic Modeling

Real-world physical interactions span diverse regimes, from rigid to non-rigid dynamics and from prehensile to non-prehensile manipulation. These regimes impose distinct kinematic and kinetic constraints, making a single monolithic transformer difficult to learn and optimize. Yet, we observe that the interaction regime is often straightforward to infer from a short history of states and actions. Motivated by this, we adopt a Mixture-of-Experts architecture that identifies motion patterns and routes to specialized dynamics decoders for multi-physics modeling (Fig. 2).

Given the current and historical scene states and actions represented as point clouds, we encode them using a scene encoder and an action encoder to obtain point *embeddings*. The embeddings are fused into a unified representation $\mathbf{z}_t$, serving as shared conditioning for both the gating function and the expert dynamics models. We predict a one-hot routing vector via a gating function $g(\cdot)$:

$$\mathbf{g}_t = g_\theta(\mathbf{z}_t); \text{ where } \mathbf{g}_t \in \{0, 1\}^K, \sum_{k=1}^K g_t^{(k)} = 1, \quad (2)$$

where K is the number of experts. The router is re-evaluated at each rollout step, allowing adaptation to changes in the interaction regime. This hard routing encourages clear expert specialization and reduces interference across heterogeneous physical regimes.

Each expert dynamics model f_k takes the same shared embedding $\mathbf{z}_t$ as input and predicts the next H -step scene states. The overall MoE prediction is $\mathbf{s}_{t+1:t+H} = \sum_{k=1}^K g_t^{(k)} f_k(\mathbf{z}_t)$, which is equivalent to activating only the selected expert under one-hot routing.

3.3 Physics-Aware Expert Design

We group the experts into two material classes, rigid and deformable, and further specialize them by interaction regime. Our model uses three experts: (i) *rigid non-prehensile*, (ii) *deformable*

non-prehensile, and (iii) *deformable prehensile*. Each expert is first trained on its corresponding simulation-domain data and then jointly fine-tuned on mixed real and simulated data. We omit rigid prehensile interactions, since rigid grasping has limited physical uncertainty when slip is negligible and can be well modeled by existing world models.

Rigid Expert. For rigid objects, motion is fully characterized by a global 6-DoF rigid transform $(\mathbf{R}_\tau, \mathbf{t}_\tau) \in \text{SE}(3)$. Accordingly, the rigid expert predicts a sequence of transforms via an internal head $(\omega_{t+1:t+H}, \mathbf{t}_{t+1:t+H}) = h_{\text{rigid}}(\mathbf{z}_t)$, where $\omega_t \in \mathbb{R}^6$ is the 6D continuous rotation representation (converted to $\mathbf{R}_t \in \text{SO}(3)$) and $\mathbf{t}_t \in \mathbb{R}^3$ is translation. The expert then outputs future scene point clouds by rigid warping $f_{\text{rigid}}(\mathbf{z}_t; \tau) = \mathbf{R}_{t+1:t+H} \cdot \mathbf{s}_t + \mathbf{t}_{t+1:t+H}$. This parameterization enforces rigidity and global coherence, reducing the hypothesis space and improving stability compared to free-form point decoding.

Deformable Experts. Deformable objects require free-form geometry updates beyond global rigid warping. We therefore instantiate *two* deformable experts with identical architecture but separate parameters and training: one specialized for prehensile interactions and one for non-prehensile interactions. The prehensile expert specializes in force-induced shape changes under grasping (e.g., squeezing and stretching), whereas the non-prehensile expert focuses on contact-induced deformations from pushing and sliding interactions. For simplicity, we omit explicit expert indices in the notation. Given the shared embedding $\mathbf{z}_t$, the selected deformable expert predicts point-wise $\Delta \mathbf{s}_{t+1:t+H} = h_{\text{def}}(\mathbf{z}_t)$, and outputs future scene states by applying the displacement $f_{\text{def}}(\mathbf{z}_t) = \mathbf{s}_{t:t+H-1} + \Delta \mathbf{s}_{t+1:t+H}$.

3.4 Training and Inference

Training During training, we adopt a self-rolling strategy [63] to match inference-time behavior and mitigate distribution shift. Starting from the current scene state $\mathbf{s}_t$ (and the history buffer), the model predicts the next-step scene $\hat{\mathbf{s}}_{t+1}$. We then feed $\hat{\mathbf{s}}_{t+1}$ back as input to predict subsequent steps, rolling out autoregressively over a horizon of H steps. Following prior work, gradients are detached through the rolled-in predictions, and losses are computed over the entire rollout horizon.

We optimize a combined objective consisting of a reconstruction loss and an expert routing loss:

$$\mathcal{L} = \sum_{\ell=1}^H \|\mathbf{s}_{t+\ell} - \mathbf{s}_{t+\ell}^{\text{gt}}\|_2^2 + \lambda \sum_{\ell=1}^H \text{CE}(\mathbf{g}_{t+\ell}, \mathbf{g}_{t+\ell}^{\text{gt}}), \quad (3)$$

where $\mathbf{s}^{\text{gt}}$ denotes ground-truth scene point clouds, $\hat{\mathbf{g}}$ is the predicted one-hot routing distribution from the gating function $g(\cdot)$, and $\mathbf{g}^{\text{gt}}$ is the supervision signal for expert selection. We first warm up the gating network and experts separately on simulation data, and then jointly fine-tune the full model on mixed real-world data.

Inference At inference time, the model performs fully autoregressive rollout. Given the observed initial scene and a history buffer initialized from past states/actions, we iteratively predict $\mathbf{s}_{t+1}, \mathbf{s}_{t+2}, \dots$. After each step, the predicted state is appended to the history buffer and used as input for the next prediction. The procedure repeats until reaching the desired prediction horizon.

3.5 Planning with World Model

We use the learned world model as the predictive dynamics in an Model Predictive Control (MPC) loop. At each control step, given the current observed state $\mathbf{s}_t$ (scene point cloud) and history, we sample a set of candidate action sequences $\{\mathbf{a}_{t:t+H-1}^{(j)}\}_{j=1}^J$ over a planning horizon H and roll out the world model to obtain predicted trajectories $\{\mathbf{s}_{t+1:t+H}^{(j)}\}$. We evaluate each trajectory with a task-specific cost that measures progress toward a goal state $\mathbf{s}_{\text{goal}}$ (e.g., distance between the predicted terminal scene and the goal), and select the lowest-cost plan j^* . We then execute only the first

Table 2: **Dynamic model comparison** across interaction types and physics. We report geometric accuracy (MSE, CD) and material structural metrics (rigidity or Laplacian). Privileged experts (gray) are domain-specific and assume known interaction types at test time, while our Mixture-of-Experts model handles diverse interactions in a unified framework with consistently strong performance.

Method	Rigid-body Non-Prehensile			Deformable Non-Prehensile			Deformable Prehensile		
	RMSE ↓	CD ↓	Rigidity ↓	RMSE ↓	CD ↓	Laplacian ↓	RMSE ↓	CD ↓	Laplacian ↓
AdaptiGraph	0.028	0.084	0.0001	0.031	0.052	0.0068	0.047	0.143	0.0092
ParticleFormer	0.054	0.082	0.0005	0.061	0.079	0.0072	0.058	0.075	0.0058
Privileged Expert	0.021	0.048	0.0	0.018	<u>0.032</u>	<u>0.0027</u>	0.029	0.038	0.0068
Ours (MoE)	<u>0.022</u>	0.048	0.0	<u>0.024</u>	0.027	0.0020	0.026	<u>0.039</u>	0.0050

Table 3: **Ablation of MoRE** under different interaction types and physical regimes. We evaluate the impact of expert specialization, routing strategy, and self-rollout training using geometric and structural metrics. Performance degrades when removing expert specialization, explicit routing, or self-rollout training, highlighting the importance of each component for robust dynamics prediction.

Method	Rigid-body Non-Prehensile			Deformable Non-Prehensile			Deformable Prehensile		
	RMSE ↓	CD ↓	Rigidity ↓	RMSE ↓	CD ↓	Laplacian ↓	RMSE ↓	CD ↓	Laplacian ↓
Single Expert	0.030	0.054	0.0001	0.028	0.038	0.0038	0.037	0.047	0.0061
Implicit Router	0.026	0.051	0.0	0.032	0.033	0.0032	0.028	0.040	0.0054
Implicit MoE	0.045	0.070	0.0002	0.026	0.031	0.0032	0.032	0.039	0.0071
w/o Self-Rollout	0.031	0.057	0.0	0.035	0.039	0.0056	0.047	0.060	0.0058
Ours (Full)	0.022	0.048	0.0	0.024	0.027	0.0020	0.026	0.039	0.0050

action $a_t^{(j^*)}$, observe the next state, and re-plan at the next step. For obstacle present (e.g. multi-object manipulation), we add additional cost for collision avoidance. Other MPC details (action sampling, costs, and constraints) are provided in the following sections.

4 Experiments

4.1 Implementation Details

Dataset We use a mixed simulation-real dataset for training, covering diverse physical regimes, and interaction types. The simulation data are collected from Isaac Sim and contain approximately 20K episodes. We automatically generate diverse interactions with random pushing actions, and physical properties are randomized per episode. The real-world data contain approximately 7K episodes of DROID [64] and [12]. Additional dataset details can be found in the supplementary.

Model Architecture Our model takes a temporal sequence of point sets as input, using 8 history steps to predict a 3-step horizon. The object point cloud is represented with a variable number of points based on object size, together with 100 end-effector points. Details about network architecture can be found in the supplementary.

4.2 Tasks and Evaluation Setup

Tasks Our evaluation covers three representative manipulation regimes. In rigid non-prehensile tasks, 30 different tools from [65] are used to push or pull 300 rigid objects from [66], emphasizing object coherence and contact consistency. In deformable non-prehensile tasks, the same tools interact with 300 soft objects from [67] through pushing and pulling, requiring accurate modeling of local deformation. In deformable prehensile tasks, the robot grasps and moves deformable objects such as toys and clothes, introducing sustained contact and large non-rigid shape changes.

Metrics and Evaluation Setup We evaluate dynamic prediction with geometric accuracy and material-consistent structural metrics, as summarized in Table 2. Geometric accuracy is measured

Table 4: **Comparison with Point-World on the DROID dataset.** Our method performs better across all three cases, showing strong generalization to real-world object dynamics.

Method	Rigid-body Non-Prehensile			Deformable Non-Prehensile			Deformable Prehensile		
	RMSE ↓	CD ↓	Rigidity ↓	RMSE ↓	CD ↓	Laplacian ↓	RMSE ↓	CD ↓	Laplacian ↓
Point-world	0.031	0.054	0.0003	0.028	0.031	0.0041	0.029	0.035	0.0063
Ours	0.023	0.046	0.0	0.014	0.026	0.0022	0.021	0.031	0.0051

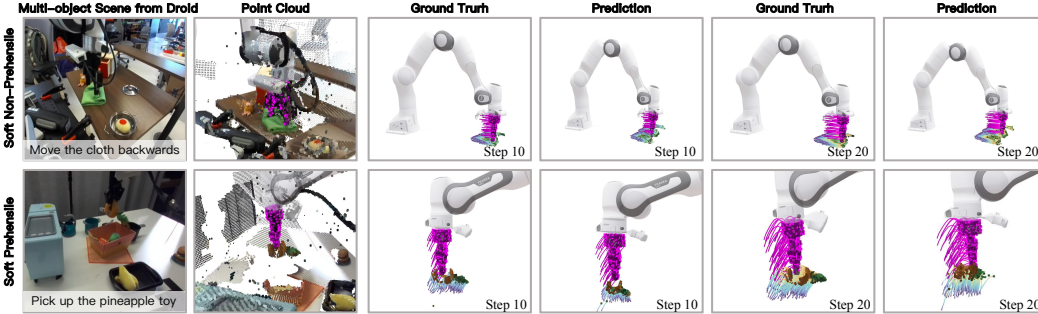


Figure 3: **Qualitative results on DROID evaluation.** We compare model predictions with ground truth on challenging DROID scenes with clutter, diverse objects, and partial observations. Even for small, low-resolution manipulated objects, our model produces accurate dynamics, demonstrating robustness to occlusion and noise.

by MSE and Chamfer Distance (CD) between predicted and ground-truth point clouds. For rigid interactions, we report rigidity error by fitting an optimal rigid transformation with ICP and measuring residual alignment error. For deformable objects, we report Laplacian error between predicted and ground-truth point clouds, where lower values indicate smoother, more plausible deformations.

4.3 Dynamic Modeling across Interactions and Physics

We compare against different baselines in Tables 2 and 4. In Table 2, we compare with Adapti-Graph [12] and ParticleFormer [15], which are neural dynamics models mainly designed for deformable object modeling, as well as privileged experts trained on specific interaction domains.

Across all regimes in Table 2, existing methods perform well only within their targeted categories and struggle to generalize across interaction types and physical properties. In contrast, our MoE model does not rely on privileged test-time interaction labels, yet achieves performance comparable to domain-specific experts across diverse physics regimes. Its shared representation and routing enable effective specialization for heterogeneous manipulation dynamics.

In Table 4, we further compare with PointWorld on DROID, using its model trained directly on DROID data. Since our goal is manipulated-object dynamics, we evaluate only moving object points rather than whole-scene flow. Our method achieves better performance across all three cases, demonstrating strong generalization to real-world object dynamics. Visualizations of different scenes and comparisons between our predictions and ground truth are shown in Fig. 3.

4.4 Ablation Study on Motion Experts and Routing

Table 3 presents an ablation study on architectural and training design choices. All ablation models are trained on the same dataset with approximately matched model capacity. Training a single expert to model all physics leads to degraded performance, struggling to capture both rigid and deformable behaviors. Implicit routing, which softly combines expert outputs without supervised binary gating, further reduces geometric accuracy and physical consistency, highlighting the importance of explicit expert selection. Implicit MoE embeds expert specialization within the transformer using a shared, displacement-based output representation, rather than employing explicit motion experts with distinct roles. This design consistently underperforms across all regimes, suggesting

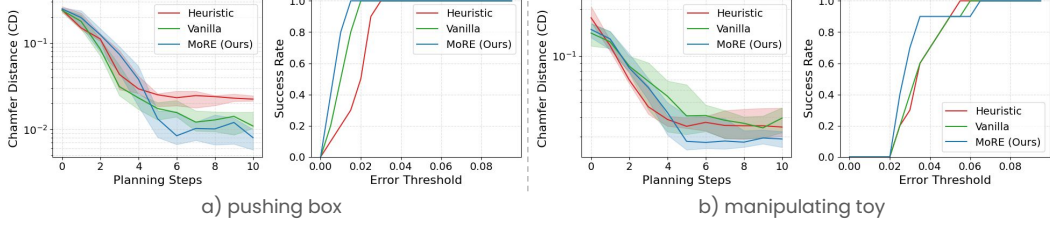


Figure 4: **Model Predictive Control (MPC) performance.** (a) Box pushing and (b) toy manipulation results. Each section details the prediction error across planning steps (left) and success rate vs. error threshold (right). MoRE achieves lower error and higher success than the baselines.

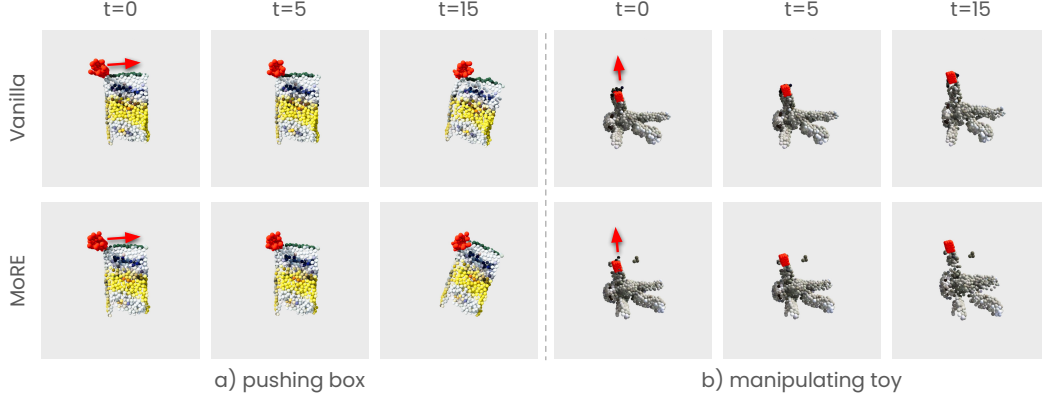


Figure 5: **Real world dynamics predictions.** We visualize predictions of point motion on real world data. The red points are the end-effector (action) points. The Vanilla model struggles to treat the box as rigid and the toy sloth as soft. In contrast, our model correctly handles the different cases.

that implicit capacity allocation alone is insufficient to capture heterogeneous dynamics. Finally, removing self-rollout training and relying solely on teacher forcing results in less stable predictions, particularly for contact-rich deformable interactions.

4.5 MPC Performance with Learned World Models

Setup We evaluate downstream control performance with MPC in real-world robot experiments, including rigid box pushing and deformable toy-sloth pose matching. Goals are specified as target object point clouds, and the MPC cost is defined as the bidirectional Chamfer Distance (CD) between predicted and target point clouds. Success is measured using a threshold on the actual CD after execution. We compare against two baselines: *Heuristic MPC*, which assumes all object points follow the end-effector translation, and *Vanilla**, a single Transformer-based world model without motion experts or routing. Additional MPC details are provided in the supplementary.

Results As shown in Fig. 4, our method consistently achieves lower CD compared to baselines. Heuristic MPC struggles to generalize across tasks due to hand-designed, task-specific objectives, while *Vanilla** suffers from limited capacity to model heterogeneous physical dynamics. In contrast, our physics-aware, expert-based 3D world model enables more accurate long-horizon prediction, translating to more efficient MPC. Prediction rollouts are visualized in Fig. 5.

5 Discussion and Limitations

We introduced MoRE, a 3D-aware robotic world model for predicting interaction dynamics across heterogeneous physical regimes. By modeling scenes and actions directly in point cloud space and decomposing dynamics into specialized motion experts, MoRE enables contact-rich reasoning and robust modeling of both rigid and deformable interactions.

Limitations Our method currently focuses on modeling the motion of a single manipulated object and does not handle simultaneous multi-object interactions. Also, it does not explicitly model fine-grained physical properties (e.g. mass, friction, elasticity), which may limit performance for visually similar objects with different internal states, such as an empty bottle versus a water-filled bottle.

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References

- [1] G. Zhou, H. Pan, Y. LeCun, and L. Pinto. Dino-wm: World models on pre-trained visual features enable zero-shot planning. *arXiv preprint arXiv:2411.04983*, 2024.
- [2] S. Zhou, Y. Du, J. Chen, Y. Li, D.-Y. Yeung, and C. Gan. Robodreamer: Learning compositional world models for robot imagination. *arXiv preprint arXiv:2404.12377*, 2024.
- [3] W. Huang, Y.-W. Chao, A. Mousavian, M.-Y. Liu, D. Fox, K. Mo, and L. Fei-Fei. Point-world: Scaling 3d world models for in-the-wild robotic manipulation. *arXiv preprint arXiv:2601.03782*, 2026.
- [4] D. Ha and J. Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2(3), 2018.
- [5] OpenAI. Video generation models as world simulators, 2024.
- [6] Y. Guo, L. X. Shi, J. Chen, and C. Finn. Ctrl-world: A controllable generative world model for robot manipulation. *arXiv preprint arXiv:2510.10125*, 2025.
- [7] P. Wu, A. Escontrela, D. Hafner, P. Abbeel, and K. Goldberg. Daydreamer: World models for physical robot learning. In *Conference on robot learning*, pages 2226–2240. PMLR, 2023.
- [8] H. Shi, H. Xu, S. Clarke, Y. Li, and J. Wu. Robocook: Long-horizon elasto-plastic object manipulation with diverse tools. *arXiv preprint arXiv:2306.14447*, 2023.
- [9] R. W. Sumner, J. Schmid, and M. Pauly. Embedded deformation for shape manipulation. In *ACM siggraph 2007 papers*, pages 80–es. 2007.
- [10] X. Lin, Y. Wang, J. Olkin, and D. Held. Softgym: Benchmarking deep reinforcement learning for deformable object manipulation. In *Conference on Robot Learning*, pages 432–448. PMLR, 2021.
- [11] K. Zhang, B. Li, K. Hauser, and Y. Li. Particle-grid neural dynamics for learning deformable object models from rgb-d videos. *arXiv preprint arXiv:2506.15680*, 2025.
- [12] K. Zhang, B. Li, K. Hauser, and Y. Li. Adaptigraph: Material-adaptive graph-based neural dynamics for robotic manipulation. *arXiv preprint arXiv:2407.07889*, 2024.
- [13] H. Jiang, H.-Y. Hsu, K. Zhang, H.-N. Yu, S. Wang, and Y. Li. Phystwin: Physics-informed reconstruction and simulation of deformable objects from videos. *arXiv preprint arXiv:2503.17973*, 2025.
- [14] T. Xie, Z. Zong, Y. Qiu, X. Li, Y. Feng, Y. Yang, and C. Jiang. Physgaussian: Physics-integrated 3d gaussians for generative dynamics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4389–4398, 2024.

- [15] S. Huang, Q. Chen, X. Zhang, J. Sun, and M. Schwager. Particleformer: A 3d point cloud world model for multi-object, multi-material robotic manipulation. *arXiv preprint arXiv:2506.23126*, 2025.
- [16] Y. Huang, C. Agia, J. Wu, T. Hermans, and J. Bohg. Points2plans: From point clouds to long-horizon plans with composable relational dynamics. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1208–1216. IEEE, 2025.
- [17] T. Du, K. Wu, P. Ma, S. Wah, A. Spielberg, D. Rus, and W. Matusik. Diffpd: Differentiable projective dynamics. *ACM Transactions on Graphics (ToG)*, 41(2):1–21, 2021.
- [18] R. Zheng, J. Wang, S. Reed, J. Bjorck, Y. Fang, F. Hu, J. Jang, K. Kundalia, Z. Lin, L. Magne, et al. Flare: Robot learning with implicit world modeling. *arXiv preprint arXiv:2505.15659*, 2025.
- [19] Z. Chen, K. Yi, Y. Li, M. Ding, A. Torralba, J. B. Tenenbaum, and C. Gan. Comphy: Compositional physical reasoning of objects and events from videos. *arXiv preprint arXiv:2205.01089*, 2022.
- [20] Y. Li, J. Wu, R. Tedrake, J. B. Tenenbaum, and A. Torralba. Learning particle dynamics for manipulating rigid bodies, deformable objects, and fluids. *arXiv preprint arXiv:1810.01566*, 2018.
- [21] X. Lin, Y. Wang, Z. Huang, and D. Held. Learning visible connectivity dynamics for cloth smoothing. In *Conference on Robot Learning*, pages 256–266. PMLR, 2022.
- [22] P. Ma, P. Y. Chen, B. Deng, J. B. Tenenbaum, T. Du, C. Gan, and W. Matusik. Learning neural constitutive laws from motion observations for generalizable pde dynamics. In *International Conference on Machine Learning*, pages 23279–23300. PMLR, 2023.
- [23] Z. Xu, J. Wu, A. Zeng, J. B. Tenenbaum, and S. Song. Densephysnet: Learning dense physical object representations via multi-step dynamic interactions. *arXiv preprint arXiv:1906.03853*, 2019.
- [24] P. Battaglia, R. Pascanu, M. Lai, D. Jimenez Rezende, et al. Interaction networks for learning about objects, relations and physics. *Advances in neural information processing systems*, 29, 2016.
- [25] A. Sanchez-Gonzalez, J. Godwin, T. Pfaff, R. Ying, J. Leskovec, and P. Battaglia. Learning to simulate complex physics with graph networks. In *International conference on machine learning*, pages 8459–8468. PMLR, 2020.
- [26] H. Shi, H. Xu, Z. Huang, Y. Li, and J. Wu. Robocraft: Learning to see, simulate, and shape elasto-plastic objects in 3d with graph networks. *The International Journal of Robotics Research*, 43(4):533–549, 2024.
- [27] A. Hu, L. Russell, H. Yeo, Z. Murez, G. Fedoseev, A. Kendall, J. Shotton, and G. Corrado. Gaia-1: A generative world model for autonomous driving. *arXiv preprint arXiv:2309.17080*, 2023.
- [28] S. Liu, X. Ren, T. Shen, H. Ling, S. Gupta, S. Wang, S. Fidler, and J. Gao. Moright: Motion control done right. *arXiv preprint arXiv:2604.07348*, 2026.
- [29] J. Robine, M. Höftmann, T. Uelwer, and S. Harmeling. Transformer-based world models are happy with 100k interactions. *arXiv preprint arXiv:2303.07109*, 2023.
- [30] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

- [31] W. Fedus, B. Zoph, and N. Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.
- [32] E. Todorov, T. Erez, and Y. Tassa. Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ international conference on intelligent robots and systems*, pages 5026–5033. IEEE, 2012.
- [33] Y. Qiao, J. Liang, V. Koltun, and M. Lin. Differentiable simulation of soft multi-body systems. *Advances in Neural Information Processing Systems*, 34:17123–17135, 2021.
- [34] S. Liu, Z. Ren, S. Gupta, and S. Wang. Physgen: Rigid-body physics-grounded image-to-video generation. In *European Conference on Computer Vision*, pages 360–378. Springer, 2024.
- [35] Z. Ferguson, M. Li, T. Schneider, F. Gil-Ureta, T. Langlois, C. Jiang, D. Zorin, D. M. Kaufman, and D. Panozzo. Intersection-free rigid body dynamics. *ACM Transactions on Graphics*, 40(4), 2021.
- [36] L. Lan, D. M. Kaufman, M. Li, C. Jiang, and Y. Yang. Affine body dynamics: Fast, stable & intersection-free simulation of stiff materials. *arXiv preprint arXiv:2201.10022*, 2022.
- [37] D. Baraff. An introduction to physically based modeling: rigid body simulation i—unconstrained rigid body dynamics. *SIGGRAPH course notes*, 82, 1997.
- [38] D. Baraff. Physically based modeling: Rigid body simulation. *Siggraph Course Notes, Acm Siggraph*, 2(1):2–1, 2001.
- [39] E. Coumans and Y. Bai. Pybullet, a python module for physics simulation for games, robotics and machine learning. <http://pybullet.org>, 2016–2021.
- [40] J. J. Commandeur and S. J. Koopman. *An introduction to state space time series analysis*. Oxford university press, 2007.
- [41] B. Chen, H. Jiang, S. Liu, S. Gupta, Y. Li, H. Zhao, and S. Wang. Physgen3d: Crafting a miniature interactive world from a single image. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 6178–6189, 2025.
- [42] A. De Vaucorbeil, V. P. Nguyen, S. Sinaie, and J. Y. Wu. Material point method after 25 years: Theory, implementation, and applications. *Advances in applied mechanics*, 53:185–398, 2020.
- [43] J. C. Simo and T. J. Hughes. *Computational inelasticity*. Springer, 1998.
- [44] Y. Hu, Y. Fang, Z. Ge, Z. Qu, Y. Zhu, A. Pradhana, and C. Jiang. A moving least squares material point method with displacement discontinuity and two-way rigid body coupling. *ACM Transactions on Graphics (TOG)*, 37(4):1–14, 2018.
- [45] A. Stomakhin, C. Schroeder, L. Chai, J. Teran, and A. Selle. A material point method for snow simulation. *ACM Transactions on Graphics (TOG)*, 32(4):1–10, 2013.
- [46] M. Müller, B. Heidelberger, M. Hennix, and J. Ratcliff. Position based dynamics. *Journal of Visual Communication and Image Representation*, 18(2):109–118, 2007.
- [47] F. Liu, E. Su, J. Lu, M. Li, and M. C. Yip. Robotic manipulation of deformable rope-like objects using differentiable compliant position-based dynamics. *IEEE Robotics and Automation Letters*, 8(7):3964–3971, 2023.
- [48] B. Ai, S. Tian, H. Shi, Y. Wang, T. Pfaff, C. Tan, H. I. Christensen, H. Su, J. Wu, and Y. Li. A review of learning-based dynamics models for robotic manipulation. *Science Robotics*, 10(106):eadt1497, 2025.

- [49] L. Zhong, H.-X. Yu, J. Wu, and Y. Li. Reconstruction and simulation of elastic objects with spring-mass 3d gaussians. In *European Conference on Computer Vision*, pages 407–423. Springer, 2024.
- [50] T. Zhang, H.-X. Yu, R. Wu, B. Y. Feng, C. Zheng, N. Snavely, J. Wu, and W. T. Freeman. PhysDreamer: Physics-based interaction with 3d objects via video generation. In *European Conference on Computer Vision*. Springer, 2024.
- [51] H.-y. Chen, E. Tretschk, T. Stuyck, P. Kadlec, L. Kavan, E. Vouga, and C. Lassner. Virtual elastic objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15827–15837, 2022.
- [52] S. Le Cleac’h, H.-X. Yu, M. Guo, T. Howell, R. Gao, J. Wu, Z. Manchester, and M. Schwager. Differentiable physics simulation of dynamics-augmented neural objects. *IEEE Robotics and Automation Letters*, 8(5):2780–2787, 2023.
- [53] T. Pfaff, M. Fortunato, A. Sanchez-Gonzalez, and P. Battaglia. Learning mesh-based simulation with graph networks. In *International conference on learning representations*, 2020.
- [54] S. Guan, H. Deng, Y. Wang, and X. Yang. Neurofluid: Fluid dynamics grounding with particle-driven neural radiance fields. In *International conference on machine learning*, pages 7919–7929. PMLR, 2022.
- [55] A. Bardes, Q. Garrido, J. Ponce, X. Chen, M. Rabbat, Y. LeCun, M. Assran, and N. Ballas. Revisiting feature prediction for learning visual representations from video. *arXiv preprint arXiv:2404.08471*, 2024.
- [56] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.
- [57] N. Hansen, H. Su, and X. Wang. Td-mpc2: Scalable, robust world models for continuous control. *arXiv preprint arXiv:2310.16828*, 2023.
- [58] R. Mendonca, S. Bahl, and D. Pathak. Structured world models from human videos. *arXiv preprint arXiv:2308.10901*, 2023.
- [59] P.-C. Ko, J. Mao, Y. Du, S.-H. Sun, and J. B. Tenenbaum. Learning to act from actionless videos through dense correspondences. *arXiv preprint arXiv:2310.08576*, 2023.
- [60] S. Levine, C. Finn, T. Darrell, and P. Abbeel. End-to-end training of deep visuomotor policies. *Journal of Machine Learning Research*, 17(39):1–40, 2016.
- [61] M. Shridhar, L. Manuelli, and D. Fox. Perceiver-actor: A multi-task transformer for robotic manipulation. In *Conference on Robot Learning*, pages 785–799. PMLR, 2023.
- [62] N. Di Palo and E. Johns. Keypoint action tokens enable in-context imitation learning in robotics. *arXiv preprint arXiv:2403.19578*, 2024.
- [63] X. Huang, Z. Li, G. He, M. Zhou, and E. Shechtman. Self forcing: Bridging the train-test gap in autoregressive video diffusion. *arXiv preprint arXiv:2506.08009*, 2025.
- [64] A. Khazatsky, K. Pertsch, S. Nair, A. Balakrishna, S. Dasari, S. Karamcheti, S. Nasiriany, M. K. Srirama, L. Y. Chen, K. Ellis, et al. Droid: A large-scale in-the-wild robot manipulation dataset. *arXiv preprint arXiv:2403.12945*, 2024.
- [65] C. Lin, H. Yuan, Y. Wang, X. Qiu, T.-H. Wang, M. Guo, B. Wang, Y. Narang, D. Fox, and C. Gan. Robotsmith: Generative robotic tool design for acquisition of complex manipulation skills. *arXiv preprint arXiv:2506.14763*, 2025.

- [66] T. Chen, Z. Chen, B. Chen, Z. Cai, Y. Liu, Z. Li, Q. Liang, X. Lin, Y. Ge, Z. Gu, W. Deng, Y. Guo, T. Nian, X. Xie, Q. Chen, K. Su, T. Xu, G. Liu, M. Hu, H. ang Gao, K. Wang, Z. Liang, Y. Qin, X. Yang, P. Luo, and Y. Mu. Robotwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation, 2025. URL <https://arxiv.org/abs/2506.18088>.
- [67] S. Stojanov, A. Thai, and J. M. Rehg. Using shape to categorize: Low-shot learning with an explicit shape bias. 2021.
- [68] M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Muñoz, X. Yao, R. Zurbrugg, N. Rudin, L. Wawrzyniak, M. Rakhsha, A. Denzler, E. Heiden, A. Borovicka, O. Ahmed, I. Akinola, A. Anwar, M. T. Carlson, J. Y. Feng, A. Garg, R. Gasoto, L. Gulich, Y. Guo, M. Gussert, A. Hansen, M. Kulkarni, C. Li, W. Liu, V. Makoviychuk, G. Malczyk, H. Mazhar, M. Moghani, A. Murali, M. Noseworthy, A. Poddubny, N. Ratliff, W. Rehberg, C. Schwarke, R. Singh, J. L. Smith, B. Tang, R. Thaker, M. Trepte, K. Van Wyk, F. Yu, A. Millane, V. Ramasamy, R. Steiner, S. Subramanian, C. Volk, C. Chen, N. Jawale, A. V. Kuruttukulam, M. A. Lin, A. Mandlekar, K. Patzwaldt, J. Welsh, J.-F. Lafleche, N. Moënnelocoz, S. Park, R. Stepinski, D. Van Gelder, C. Amevor, J. Carius, J. Chang, A. He Chen, P. d. H. Ciechowski, G. Daviet, M. Mohajerani, J. von Muralt, V. Reutsky, M. Sauter, S. Schirm, E. L. Shi, P. Terdiman, K. Vilella, T. Widmer, G. Yeoman, T. Chen, S. Grizan, C. Li, L. Li, C. Smith, R. Wiltz, K. Alexis, Y. Chang, L. J. Fan, F. Farshidian, A. Handa, S. Huang, M. Hutter, Y. Narang, S. Pouya, S. Sheng, Y. Zhu, M. Macklin, A. Moravanszky, P. Reist, Y. Guo, D. Hoeller, and G. State. Isaac Lab - A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning. *arXiv preprint arXiv:2511.04831*, 2025. doi:10.48550/arXiv.2511.04831. URL <https://arxiv.org/abs/2511.04831>.
- [69] A. Khazatsky, K. Pertsch, S. Nair, A. Balakrishna, S. Dasari, S. Karamcheti, S. Nasiriany, M. K. Srirama, L. Y. Chen, K. Ellis, P. D. Fagan, J. Hejna, M. Itkina, M. Lepert, Y. J. Ma, P. T. Miller, J. Wu, S. Belkhale, S. Dass, H. Ha, A. Jain, A. Lee, Y. Lee, M. Memmel, S. Park, I. Radosavovic, K. Wang, A. Zhan, K. Black, C. Chi, K. B. Hatch, S. Lin, J. Lu, J. Mercat, A. Rehman, P. R. Sanketi, A. Sharma, C. Simpson, Q. Vuong, H. R. Walke, B. Wulfe, T. Xiao, J. H. Yang, A. Yavary, T. Z. Zhao, C. Agia, R. Baijal, M. G. Castro, D. Chen, Q. Chen, T. Chung, J. Drake, E. P. Foster, J. Gao, V. Guizilini, D. A. Herrera, M. Heo, K. Hsu, J. Hu, M. Z. Irshad, D. Jackson, C. Le, Y. Li, K. Lin, R. Lin, Z. Ma, A. Maddukuri, S. Mirchandani, D. Morton, T. Nguyen, A. O’Neill, R. Scalise, D. Seale, V. Son, S. Tian, E. Tran, A. E. Wang, Y. Wu, A. Xie, J. Yang, P. Yin, Y. Zhang, O. Bastani, G. Berseth, J. Bohg, K. Goldberg, A. Gupta, A. Gupta, D. Jayaraman, J. J. Lim, J. Malik, R. Martín-Martín, S. Ramamoorthy, D. Sadigh, S. Song, J. Wu, M. C. Yip, Y. Zhu, T. Kollar, S. Levine, and C. Finn. Droid: A large-scale in-the-wild robot manipulation dataset, 2025. URL <https://arxiv.org/abs/2403.12945>.
- [70] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- [71] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, et al. Sam 2: Segment anything in images and videos. In *International Conference on Learning Representations*, volume 2025, pages 28085–28128, 2025.
- [72] J. Zhou, J. Wang, B. Ma, Y.-S. Liu, T. Huang, and X. Wang. Uni3d: Exploring unified 3d representation at scale. *arXiv preprint arXiv:2310.06773*, 2023.
- [73] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang, J. Lei, T. Ma, B. Guo, A. Kalla, M. Marks, J. Greer, M. Wang, P. Sun, R. Rädle, T. Afouras, E. Mavroudi, K. Xu, T.-H. Wu, Y. Zhou, L. Momeni, R. Hazra, S. Ding, S. Vaze, F. Porcher, F. Li, S. Li, A. Kamath, H. K. Cheng, P. Dollár, N. Ravi, K. Saenko, P. Zhang, and C. Feichtenhofer. Sam 3: Segment anything with concepts, 2025. URL <https://arxiv.org/abs/2511.16719>.

- [74] Q.-Y. Zhou, J. Park, and V. Koltun. Open3D: A modern library for 3D data processing. *arXiv:1801.09847*, 2018.

Appendix

The supplementary material provides implementation details of dataset curation, model architecture, MPC pipeline, and more qualitative results of dynamic rollout prediction and real-world results. Project Page: <https://more-robo.github.io/>.

A Implementation details

A.1 Dataset Curation

Our training dataset is constructed by mixing simulated and real-world data, with a fixed mixture ratio of 1:1 throughout training. The two data sources are curated under a unified 3D point-cloud representation, enabling the model to benefit from both the scale and controllability of simulation and the realism of real-world observations.

A.1.1 Simulation Data Generation

The simulated portion is collected using Isaac Lab [68], leveraging its large-scale parallel simulation capability to efficiently generate diverse and controllable object–tool interaction trajectories.

Assets The simulation dataset includes three categories of assets: rigid objects, deformable objects, and tools. The rigid-object set contains approximately 300 objects spanning 130 categories, sourced from RoboTwin [66]. The deformable-object set includes around 300 objects across 100 categories, adapted from [67]. The tool set consists of approximately 30 objects from RoboSmith [65], functionally categorized into pushing, holding, lifting, and flattening tools, covering a broad range of interaction primitives.

Interaction generation Interaction data collected in simulation focus on non-prehensile manipulation. To ensure diverse and physically plausible interactions, we first uniformly sample contact points on the object surface and use the corresponding surface normal information provided by the simulator to parameterize action directions. Based on this procedure, we construct two primary non-prehensile action types: push, which applies outward forces to displace the object, and reach, which induces inward pulling interactions. This design enables systematic coverage of contact configurations while avoiding hand-crafted action heuristics.

Data preprocessing During data collection, we record the full mesh states of both objects and tools at each simulation time step, including rigid-body transformations and per-vertex deformations for soft objects. From each mesh, we uniformly sample surface points and apply farthest point sampling (FPS) to obtain point clouds with well-distributed spatial coverage. The number of sampled object points is determined adaptively according to the object size, ranging from 150 to 512 points. These sampled point clouds provide consistent point correspondences across time and are used to construct object state sequences and tool action sequences as inputs to the world model.

Domain randomization To improve generalization and reduce the sim-to-real gap, we apply extensive domain randomization during simulation. Specifically, we randomize physical properties such as mass and friction coefficients for rigid objects, as well as material parameters, such as Young’s modulus, for deformable objects. In addition, to model actuation uncertainty commonly observed in real robotic systems, we inject Gaussian noise into each action at every time step.

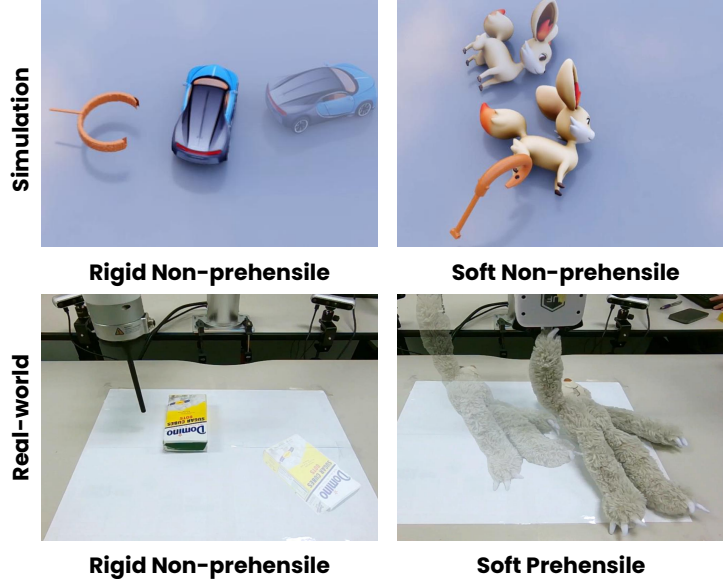


Figure 6: **Benchmark tasks visualization:** Simulation and real-world benchmarks covering rigid and soft objects under non-prehensile and prehensile interactions, designed to evaluate generalization across diverse physical regimes.

Table 5: **Gating Network Accuracy.** Expert selection accuracy (%) evaluated over 50 random interactions per task category. The gating network maintains consistently high accuracy across diverse regimes, indicating reliable expert routing for different physical dynamics.

	Push-X	Manipulate Toy	Soft Grasp	Overall
Acc.	95.4%	94.9%	99.8%	96.7%

A.1.2 Real-World Data Processing

The real-world portion is curated from two main sources: the PGND dataset [11] and the DROID dataset [69].

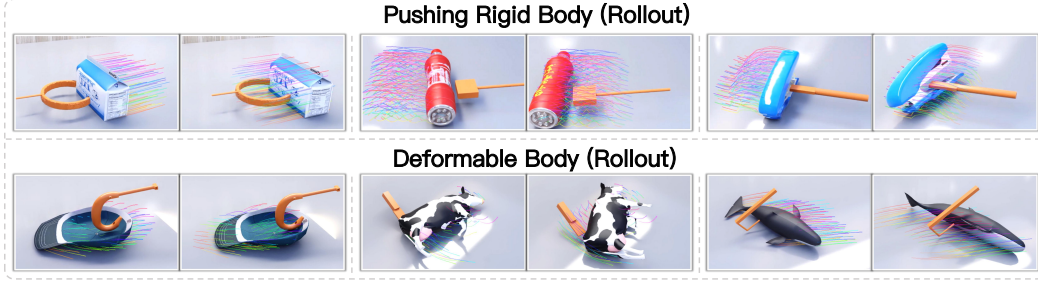
The PGND dataset provides real object-robot interaction sequences with reconstructed 3D point clouds over time and can be directly incorporated into our training pipeline.

For DROID, we follow the PointWorld [3] processing pipeline to obtain a dense 3D flow at the scene-level. Since our world model is object-centric, we further process each DROID episode to extract the target object motion. For each episode video, we first use Qwen3 [70] to identify the end-effector, the object interacted with, and the interaction type. We then apply SAM2 [71] to segment the object specified by the language-based recognition result. When multiple candidate masks are produced, we select the mask closest to the end-effector as the target object. The selected mask is finally used to filter the scene-level 3D flow, yielding the object-centric target flow used for training.

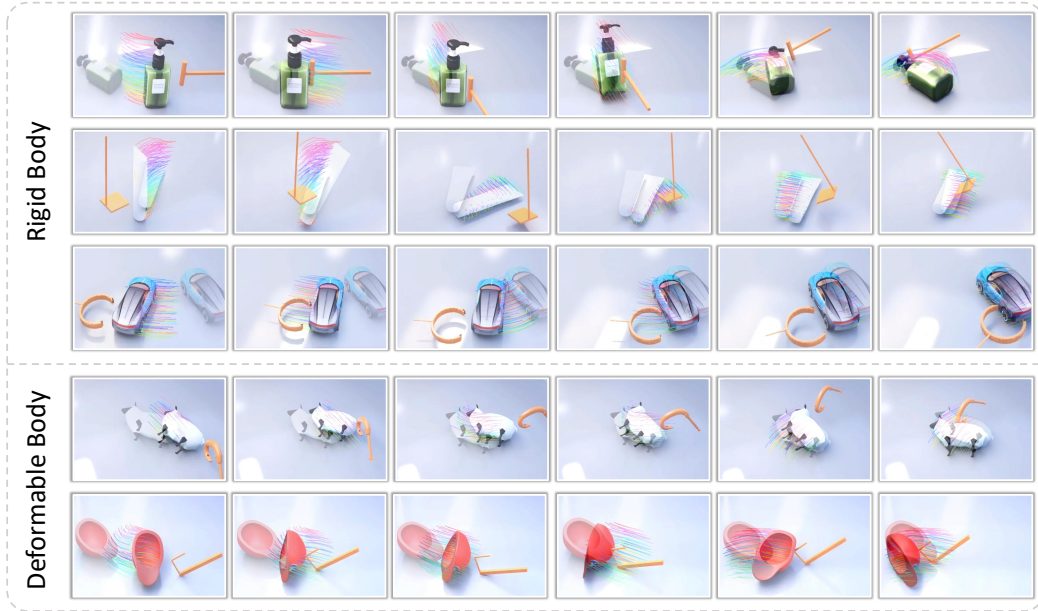
By unifying simulated and real data under a common 3D point-cloud representation and training with a balanced mixture, the model benefits from both the scale and controllability of simulation and the realism and noise characteristics of real-world observations.

A.2 Model Architecture

Backbone The model consists of two main components: a static geometry encoder and a motion encoder with a mixture-of-experts (MoE) design. The static encoder processes the current object point cloud using a frozen Uni3D [72] backbone, which encodes per-point geometric features.



(a) Dynamic prediction in simulation. **Row 1:** rigid non-prehensile pushing, where the model accurately simulates the rolling behavior of bottles under end-effector contact. **Row 2:** deformable object interactions under non-prehensile and prehensile contact, exhibiting both local deformations and global object motion.



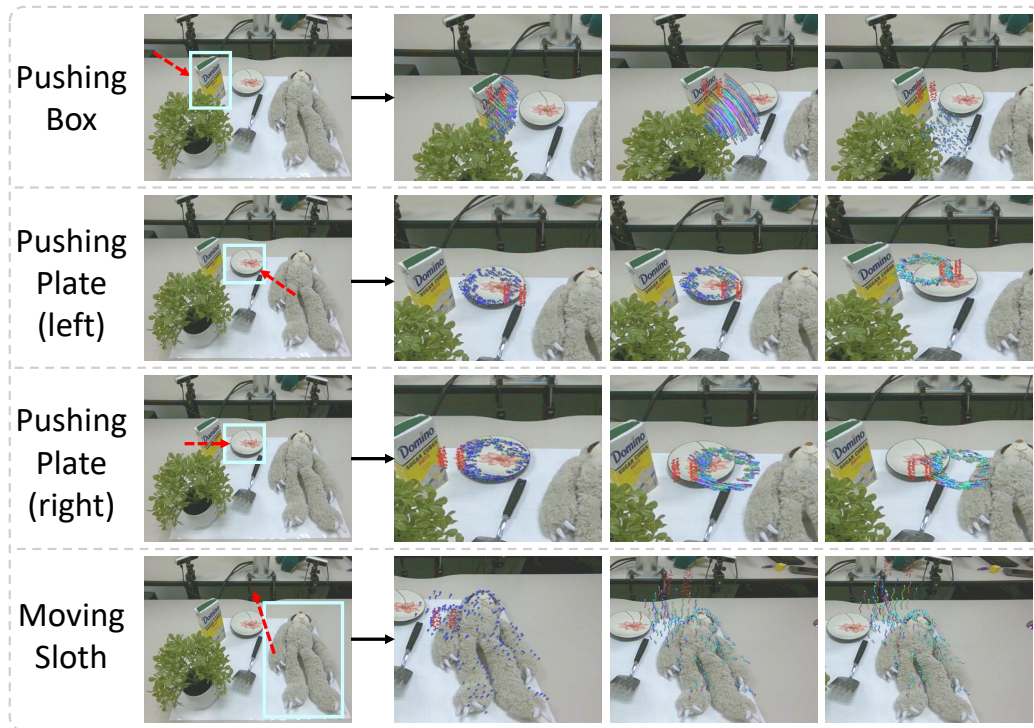
(b) MPC rollout results in simulation tasks. Qualitative simulation results of model-based control using world models trained with MoRE. The results demonstrates robust performance across diverse tools, objects, and physical regimes, including both rigid and deformable interactions.

Figure 7: **Simulation results on diverse manipulation tasks.**

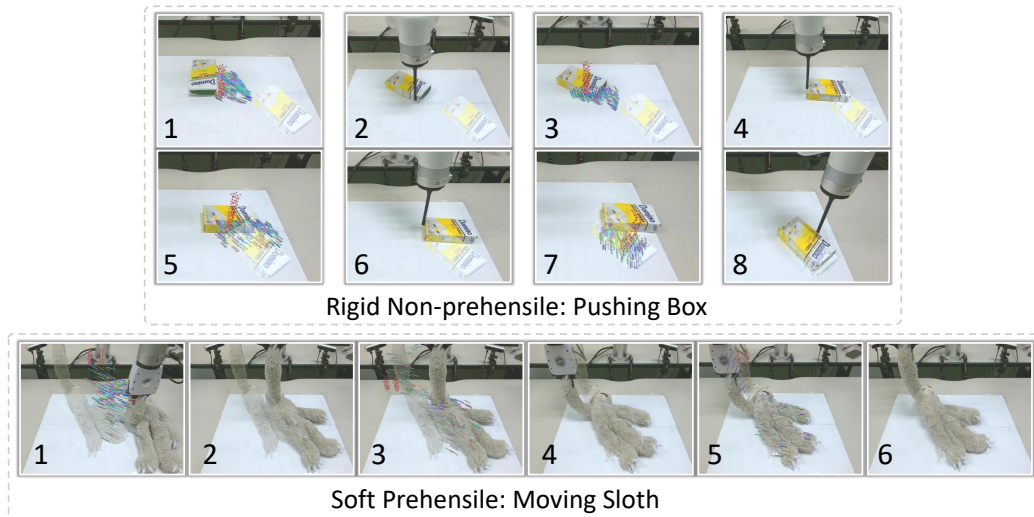
Uni3D is instantiated with grouped point tokens and a transformer-based architecture, producing 1024-dim features; its parameters are fixed during training to provide stable geometric priors. The motion encoder operates on dynamic tokens constructed from object and tool histories, and is implemented as a transformer-based network with multi-head self-attention operating on concatenated object and tool point tokens. The motion encoder outputs per-point 1024-dim motion features.

Fusion We adopt a cross-attention mechanism to fuse static geometric features with motion features, enabling the resulting latent representations to encode dynamic motion information while preserving geometric consistency throughout the interaction process.

Decoding To decode this latent representation into heterogeneous physical behaviors, we design K different motion experts. Each expert is an MLP head: the rigid expert predicts a global 6D rotation and 3D translation parameterizing an $SE(3)$ transform, while the remaining experts predict per-point 3D displacements to model deformable dynamics. Given that different physical regimes exhibit substantially different dynamics patterns, we adopt hard expert routing. Specifically, we design the gating vector as a one-hot vector, which selects a single expert at each time step within an



(a) Real-world dynamics prediction under heterogeneous physical properties, demonstrating that our unified model can accurately capture object dynamics across different material regimes.



(b) MPC rollouts on real-world manipulation tasks, showing robust control performance under sensor noise and partial observations, and highlighting the model's ability to handle contact-rich interactions across diverse physical regimes.

Figure 8: **Real-world results on diverse manipulation tasks.**

interaction sequence. The gating network is implemented as a two-layer MLP that takes the latent motion features as input and outputs expert selection logits. This design enforces explicit expert specialization and avoids blending incompatible dynamics across physical regimes.

Model forward process Given a history of object and tool point clouds, the model performs a forward prediction as follows. The current object point cloud is first encoded by the static geometry encoder to obtain per-point geometric features. Object and tool histories are flattened along the temporal dimension and embedded into dynamic tokens, which are processed by the motion encoder to produce motion features. These static and motion features are fused through cross-attention, producing geometry-aware motion representations. The gating network predicts expert weights, and each expert produces a candidate motion prediction. The rigid expert applies a global SE(3) transform to all object points, while deformable experts predict per-point displacements. The final output is selected based on the routing weights from the gating module.

A.3 MPC Inference Pipeline

Since expert routing in MoRE depends on historical object dynamics, the world model cannot directly infer the appropriate physical regime from static observations. Therefore, we adopt a two-stage MPC procedure.

Initialization We first apply a short sequence of random interactions to the object to get observable motion responses. These probing interactions provide the dynamic history required by the gating network, which then predicts the expert corresponding to the underlying physical regime.

MPC procedure Conditioning on the selected expert, we perform MPC rollouts using the learned world model. For non-prehensile manipulation, we sample 48 candidate push actions by selecting 16 contact points on the object surface and 3 pushing directions per point. For prehensile manipulation, after a grasp is established, we uniformly sample candidate tool displacements within a bounded workspace. Each candidate is represented as a short “start–end” waypoint trajectory for the tool. For each candidate, we convert the start–end tool motion into a dense action sequence by interpolating the end-effector pose over a fixed horizon. Given the current history buffer of object states and actions, we roll out the world model autoregressively for a fixed number of steps. At each step, the predicted object point cloud is appended to the history by shifting the buffer, while the corresponding future action is updated from the precomputed action sequence. After generating trajectories for all candidates, we compute the goal cost (Chamfer distance) and select the best candidate and the rollout step number (it is often better to stop early rather than complete the full action sequence). The corresponding action is executed and the system replans at the next observation.

Real-world execution In real-world experiments we normalize the coordinates by the mean of the object points so that the point clouds match the model’s training distribution. The object point cloud is obtained by fusing point clouds from four Realsense D455 cameras, each masked using SAM3 [73] with text prompts (“box”, “sloth”). Statistical outlier removal from Open3D [74] is used to remove points based on average neighbor distance. Farthest point sampling is applied in the same way as during training.

A.4 Gating Network Evaluation

In this section, we provide a detailed evaluation of the gating network. The gating network determines which expert is selected and therefore plays a critical role in the accuracy of the final predictions. We evaluate its performance on three categories of tasks: rigid non-prehensile, soft non-prehensile, and soft prehensile manipulation. For each category of tasks, we perform 50 random interaction trials and measure the expert selection accuracy. As shown in Table 5, the gating network can effectively infer the appropriate expert from historical object dynamics, enabling accurate expert routing and high-fidelity dynamics prediction.

B Additional Qualitative Results

B.1 Benchmark Task Visualization

We visualize representative samples from the four benchmark tasks used in our evaluation in Fig. 6. The tasks span rigid and soft objects under non-prehensile and prehensile interactions across both simulation and real-world settings, covering diverse contact dynamics and interactions.

B.2 Simulation Visualization

In this section, we provide additional simulation results on model-based control using world models trained with MoRE. We evaluate our approach on different non-prehensile manipulation tasks involving diverse rigid objects, deformable objects, and various tools. The qualitative results, shown in Figure 7, demonstrate that MoRE enables accurate dynamics prediction, robust handling of heterogeneous physical properties, and generalization across diverse object–tool interactions.

B.3 Real-world Visualizations

We visualize real-world model-based control results in Fig. 8. Our results show strong model performance under sensor noise and partial observations.